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#include<stdio.h>
```

```
#include<string.h>
```

```
int gcd(int a,int b){ return b==0?a:gcd(b,a%b); }
```

```
int modInv(int e,int phi){
```

```
    for(int d=1;d<phi;d++) if((d*e)%phi==1) return d;
```

```
    return -1;
```

```
}
```

```
long modExp(long base,int exp,int mod){
```

```
    long res=1;
```

```
    while(exp){
```

```
        if(exp%2) res=(res*base)%mod;
```

```
        base=(base*base)%mod;
```

```
        exp/=2;
```

```
    }
```

```
    return res;
```

```
}
```

```
int main(){
```

```
    int p,q,n,phi,e,d;
```

```
    char text[100];
```

```
    printf("Enter p q: ");
```

```
    scanf("%d%d",&p,&q);
```

```
    n=p*q;
```

```

phi=(p-1)*(q-1);

for(e=2;e<phi;e++) if(gcd(e,phi)==1) break;
d=modInv(e,phi);

printf("\nn=%d phi=%d e=%d d=%d\n",n,phi,e,d);

getchar();
printf("Enter text: ");
fgets(text,100,stdin);
text[strcspn(text,"\n")]=0;

int len=strlen(text);
long enc[100];

printf("\nEncrypted: ");
for(int i=0;i<len;i++){
    enc[i]=modExp(text[i],e,n);
    printf("%ld ",enc[i]);
}

printf("\nDecrypted: ");
for(int i=0;i<len;i++){
    printf("%c",(char)modExp(enc[i],d,n));
}

return 0;
}

```

Output:

Enter p q: 3 11

n=33 phi=20 e=3 d=7

Enter text: hi

Encrypted: 14 8

Decrypted: hi